

Collected Unit Notes

For convenience, unit notes from this module are collected here.

Unit 1: Agent-Based Systems

The focus of this unit was to examine current trends driving the growth in agent-based software solutions, including the implications of those trends to the industry overall.

I also contributed my thoughts to a collaborative discussion about the rise of agent-based systems and the benefits and drawbacks of using them in real world scenarios.

Unit 2: First Order Logic

The objectives of this unit were to review the key terms and symbols used in first order logic and to discuss their connection to natural language processing systems.

My takeaway was a solid grasp of the basics of the same.

Unit 3: Agent Architectures

The focus of this unit was a discussion of the history of agent-based development and research as well as the relative strengths of a number of different architectures.

I also summarized collaborative peer discussion about the rise of agent-based systems and the benefits and drawbacks of using them in real world scenarios.

Unit 4: Hybrid Agent Architectures

The focus of this unit was an overview of conceptual approaches for designing hybrid architectures for agent-based systems.

My takeaway was an introductory understanding of the same.

Unit 5: Agent Communication

The focus of this unit revolved around speech acts and speech act theory in the context of the ontologies used to design inter-agent communication systems.

I also contributed my thoughts to a collaborative discussion about the advantages versus the drawbacks of using KQML for messaging between agents.

Unit 6: Working Together

The main objective of this unit was to review techniques for using KQML and KIF to design messaging schemas for communication between agents.

I also completed the first major module project: a group project to develop a design specification for an automated, agent-driven content moderation system for a generic social media platform.

My project role was to write the final report. I wrote all of the project report and the first workflow process diagram; the second and third diagrams were provided by another group member.

For portfolio development, I also modeled a sample agent dialogue using KQML and KIF.

Unit 7: Natural Language Processing

The objective of this unit was to develop an introductory familiarity with the core concepts and development techniques underpinning natural language processing.

My takeaway was an introductory understanding of the same.

Unit 8: Understanding Natural Language Processing

The objectives of this unit were to examine use cases for Word2Vec models and constituency-based parse trees.

For portfolio development, I modeled three constituency-based parse trees.

Unit 9: Introduction to Adaptive Algorithms

The focus of this unit revolved around an introduction to adaptive algorithms as used in artificial neural networks and deep learning, along with a discussion of their strengths and weaknesses.

I also contributed my thoughts to a collaborative discussion about potential ethical concerns and pitfalls surrounding the content produced by generative AI platforms.

Unit 10: Deep Learning in Action

This unit continued the discussion of the same subject matter as in Unit 9.

For portfolio development, I researched a subject matter area (archaeological excavation site discovery) where deep learning could be applied to provide societal benefits that are unachievable via conventional science or engineering methods.

Unit 11: Intelligent Agents in Action

This unit focused on use cases for agent-driven systems in Industry 4.0, specifically manufacturing and FinTech.

For portfolio development, I prepared a slide presentation and live demo of an extract of the project design specification from the group project earlier in the module, an agent-driven content moderation system for a generic social media platform.

Unit 12: The Future of Intelligent Agents

The focus of this unit revolved around a discussion of the future of agent-driven software solutions development at a broad level.

I also prepared a reflective essay with my thoughts on my learning challenges and learning outcomes from the module.